SEMICONDUCTOR ELECTRONIC

1. When a semiconductor is doped its electrical conductivity :
(1) Increases
(2) Decreases in the direct ratio of the doped material
(3) Decreases in the inverse ratio of the doped material
(4) Remains unaltered

2. Which of the following semiconductor is electrically positive ?
(1) Intrinsic semiconductor
(2) P-type semiconductor
(3) N-type semiconductor
(4) None of these

3. The depletion layer of a p-n junction :
(1) is of constant width irrespective of the bias
(2) acts like an insulating zone under reverse bias
(3) has a width that increases with an increases in forward bias
(4) is depleted of ions

4. Let n_p and n_e be the numbers of holes and conduction electrons in an extrinsic semiconductor :-
(1) $n_p > n_e$ (2) $n_p = n_e$
(3) $n_p < n_e$ (4) $n_p \neq n_e$

5. Which of the following statements is **INCORRECT** :-
(1) The resistance of intrinsic semiconductor decrease with increase of temperature
(2) Doping pure Si with trivalent impurities gives P-type semiconductors
(3) The majority carriers in N-type semiconductors are holes
(4) A PN-junction can act as a semiconductor diode

6. The conductivity of a semiconductor increases with increase in temperature because
(1) number density of free current carriers increases.
(2) relaxation time increases.
(3) both number density of carriers and relaxation time increase.
(4) number density of current carriers increases, relaxation time decreases but effect of decrease in relaxation time is much less than increase in number density.

7. **Statement-1 :** The number of electrons in n-type semiconductor is higher than the number of electrons in a pure silicon semiconductor.
Statement-2 : The law of mass action is applicable to n-type semiconductors.
(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statament-2 is True; Statement-2 is NOT a correct explanation for Statement-I.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 False, Statement-2 is True.

8. In a p-n junction,
(1) new holes and conduction electrons are produced continuously throughout the material
(2) new holes and conduction electrons are produced continuously throughout the material except in the depletion region
(3) holes and conduction electrons recombine continuously throughout the material.
(4) holes and conduction electrons recombine continuously throughout the depletion region.

9. A Si and a Ge diode has identical physical dimensions. The band gap in Si is larger than that in Ge. An identical reverse bias is applied across the diodes
(1) The reverse current in Ge is larger than that in Si
(2) The reverse current in Si is larger than that in Ge
(3) The reverse current is identical in the two diodes
(4) The relative magnitude of the reverse currents cannot be determined from the given data only

10. Diffusion current in a p-n junction is greater than the drift current in magnitude :-
(1) If the junction is forward-biased
(2) If the junction is reverse-biased
(3) If the junction is unbiased
(4) In no case.

11. **Statement-1 :** Transistor can act as a switch.

Statement-2 : When the transistor is not conducting it is said to be switched off and when it is driven into saturation it is said to be switched on.

(1) Statement-I is true, Statement-II is true, Statement-II is the **correct** explanation of Statement-I

(2) Statement-I is true, Statement-II is true, Statement-II is **not** the correct explanation of Statement-I.

(3) Statement-I is **true,** Statement-II is false.

(4) Statement-I is **false,** Statement-II is true.

12. In P-type semiconductor, drift velocity of electrons compared to holes is :

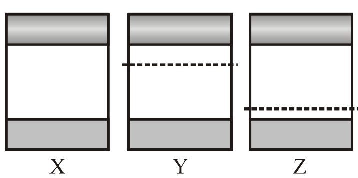
(1) More

(2) Less

(3) Equal

(4) None of these

13. The energy band diagrams for three semiconductor samples of silicon are as shown. We can then assert that :-



(1) Sample X is undoped while samples Y and Z have been doped with a trivalent and a pentavalent impurity respectively

(2) Sample X is undoped while both samples Y and Z have been doped with a pentavalent impurity

(3) Sample X has been doped with equal amounts of trivalent and pentavalent impurities while samples Y and Z are undoped

(4) Sample X is undoped while samples Y and Z have been doped with a pentavalent and a trivalent impurity respectively

14. In a transistor symbol, the arrow is drawn on ____(a)____ and it represents direction of ____(b)____, then (a) and (b) are respectively.

(1) Emitter ; hole movement

(2) Emitter ; electron movement

(3) Collector ; hole movement

(4) Base ; hole movement

15. One end of a device is connected to positive terminal and the other end to the negative terminal and the current is flowing. If the terminals of supply are interchanged, then there is approximately zero current. The device can be :-

(1) p-n junction (2) transistor

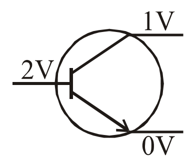
(3) capacitor (4) inductor

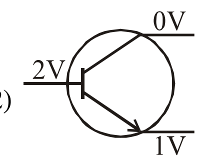
16. In a half wave rectifier if input frequency is 50 Hz then output ripple frequency will be :–

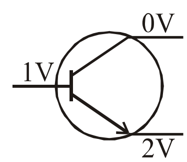
(1) 25 Hz (2) 50 Hz

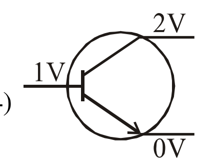
(3) 100 Hz (4) 200 Hz

17. In which of the following cases, the transistor is operating in the active region ?

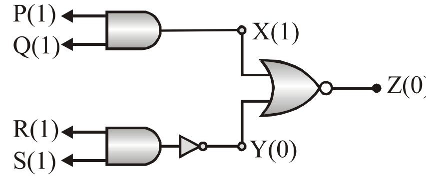
(1) 

(2) 

(3) 

(4) 

18. The circuit diagram shows a logic combination with the states of output X, Y and Z given for inputs P, Q, R and S all at state 1. When inputs P and R change to state 0 with inputs Q and S still at 1, the states of outputs X, Y and Z change to



(1) 1, 0, 0 (2) 1, 1, 1

(3) 0, 1, 0 (4) 0, 0, 1